Test-enhanced learning of clinical reasoning: a crossover randomised trial

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CONTEXT Clinical reasoning is an essential skill, the foundations of which should be acquired during undergraduate medical education. Student performance in clinical reasoning can be assessed using key feature examinations. However, within a paradigm of test-enhanced learning, such examinations may also be used to enhance long-term retention of procedural knowledge relevant to clinical reasoning.

OBJECTIVES This study tested the hypothesis that repeated testing with key feature questions is more effective than repeated case-based learning in fostering clinical reasoning.

METHODS In this randomised crossover trial, Year 4 medical students attended 10 weekly computer-based seminars during which patient case histories covering general medical conditions were displayed. The presentation format was switched between groups every week. In the control condition, students studied long case narratives. The intervention condition used the same content but augmented case presentation with a sequence of key feature questions. Using a within-subjects design, student performance on intervention and control items was assessed at 13 weeks (exit examination) and 9 months (retention test) after the first day of term.

RESULTS A total of 87 of 124 eligible students provided complete data for the longitudinal analysis (response rate: 70.2%). In the retention test, mean ± standard deviation student scores on intervention items were significantly higher than those on control items (56.0 ± 25.8% versus 48.8 ± 24.7%; $p < 0.001$). The results remained unchanged after accounting for exposure time in a linear regression analysis that also adjusted for sex and general student performance levels.

CONCLUSIONS This is the first study to demonstrate an effect of test-enhanced learning on clinical reasoning as assessed with key feature questions. In this randomised trial, repeated testing was more effective than repeated case-based learning alone. Curricular implementation of longitudinal key feature testing may considerably enhance student learning outcomes in relevant aspects of clinical medicine.

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INTRODUCTION

The overall aim of undergraduate medical education is to equip future physicians with a solid knowledge base and the essential skills required to manage their future patients. Amongst the higher-order cognitive functions that must be mastered is the act of applying knowledge to new clinical situations in order to make informed choices about diagnostic procedures and therapeutic options. These functions are usually described as 'clinical reasoning' or 'clinical cognition' and they encompass a range of strategies used to generate and test hypotheses about a disease or mechanism underlying a given patient's presenting complaint, as well as to arrive at decisions based on judgements of the prognostic significance of diagnostic tests.[1] At least two processes are involved in clinical reasoning, of which the first is intuitive (e.g. pattern recognition from first impressions) and the second is analytic in nature.[2] It has been proposed that easier cases can readily be solved by applying intuitive strategies, whereas more complex cases require more in-depth processing.[3] Clinical reasoning hinges on the availability of a solid knowledge base, deliberate practice and metacognitive processes (e.g. reflective coping[4]).

The most popular teaching formats aimed at fostering clinical reasoning are small-group sessions (e.g. in problem-based or case-based learning[5,6]) and clinical rotations. However, as clinical teachers are required to facilitate small-group teaching, this instructional format is resource-intensive. In addition, the group format is not well suited to students who learn at different speeds and is prone to the domination of discussions by particular students, which possibly hampers learning outcomes in others.[7] These problems should not occur during clinical attachments in which one-to-one teaching is available from experienced clinicians.[8] However, the content for this format cannot be standardised to the same extent as in formal teaching sessions. Learning opportunities on medical wards depend on the types of patients treated on these wards. In consequence, students may be less likely to encounter patients with specific diseases (including some for which swift decision making is crucial), thus limiting the range of content covered during clinical attachments.

One way to standardise the content covered and provide equal learning opportunities to all students is to use computer-assisted case-based learning.[9] Despite being more effective than no teaching at all, computer-based learning is in itself not generally superior to more traditional instructional formats.[10] In order to be effective, computer-based teaching interventions should be informed by theory. One recent advance in educational psychology refers to test-enhanced learning. Although it has been known for some time that summative assessments drive student learning *behaviour*,[11] the first studies assessing the impact of formative assessments on learning *processes* in medicine were only published in 2008[12] (see Roediger and Karpicke[13] for a thorough discussion of the principles underlying test-enhanced learning). These trials tended to focus on rote memory[14] rather than complex cognitive functions, and some included short follow-up periods (e.g. 1 week[15]). Although some studies used sophisticated testing methods,[16] others relied on multiple-choice questions,[17] which have been shown to be less effective in this scenario[13] and potentially even detrimental to student learning outcomes.[18] Current research on test-enhanced learning addresses mechanisms underlying the observed effects. For example, superior performance in a final test could either be the result of effortful retrieval during preceding tests ('depth of processing hypothesis'[19]) or may simply reflect a training effect if the format of these tests is similar and thus evokes identical cognitive processes ('transfer-appropriate hypothesis'[20]). In a recent study comparing these hypotheses, McConnell *et al.*[21] found that student performance was better when more demanding test formats such as short-answer questions and context-rich multiple-choice questions (as opposed to context-free questions) were used during the training phase. However, the effect seen in the final test was limited to questions that were identical to those seen during the training. From a practical perspective, medical educators need to know how repeated testing can be used both effectively and efficiently.

Given the potential lack of exposure to some relevant aspects of clinical medicine during small-group and face-to-face teaching, as well as the opportunities arising from the concept of test-enhanced learning, the present trial combined computer-based learning with key feature questions to assess whether repeated testing is more effective than repeated case-based learning in fostering clinical reasoning.[22] We hypothesised that undergraduate medical students answering key feature questions would retain more procedural knowledge over a 6 month period than students exposed to the same content without being prompted to answer key feature questions. The effect – if present – was expected to be attribu-

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table to an enhanced ability to access the relevant knowledge rather than a mere training effect regarding the handling of key feature questions.

METHODS

Study design

This was a randomised, controlled, non-blinded crossover trial involving Year 4 undergraduate medical students. At the institution at which the study was conducted, undergraduate medical education is divided into two parts. During the first 2 years, basic sciences, as well as anatomy, biochemistry, physiology and medical psychology, are taught. After passing a high-stakes examination, students progress to the clinical phase of the curriculum. This phase is made up of 21 modules, each of which lasts 2–7 weeks. Modules in the first year of the second phase are devoted to practical skills training, microbiology, pharmacology and radiology, whereas modules in the second and third years focus on signs and symptoms of specific diseases and approaches to treatment. In this study, Year 4 students enrolled in three consecutive teaching modules (cardiology and pulmonology, nephrology and rheumatology, haematology and oncology) in the winter term of 2013–2014 were eligible for study participation. Following an entry examination in the first week of term, students were stratified by sex and performance levels and randomised to one of two groups. Students in both groups attended weekly computer-assisted learning sessions ('e-seminars'; maximum duration: 45 minutes) during which they were presented with four patient case histories related to teaching content of the previous week (Table 1). Case content was identical for all students, but the format varied each week between the 'study-only cases' and the 'test cases' in that the latter also involved prompts to answer five key feature questions per case. These questions are referred to as 'items'; all analyses were performed at the item level (and not at the level of complete cases). Group A began with test cases (Fig. 1, black stripes) and Group B began with study-only cases (Fig. 1, grey

Table 1 Teaching and e-seminar content

	Content covered		E-seminar format	
Week	**Formal teaching**	**E-seminars**	**Group A**	**Group B**
1	Coronary artery disease	All (entry examination)	T	T
2	Heart failure	Coronary artery disease	T	S
3	Valvular disease	Heart failure	S	T
4	Arrhythmias	Valvular disease, coronary artery disease	T	S
5	Respiratory disease	Arrhythmias, heart failure	S	T
6	Peripheral artery disease, pulmonary embolism, myocarditis, pericarditis	Respiratory disease, valvular disease, coronary artery disease	T	S
7	Nephrotic syndrome	Peripheral artery disease, pulmonary embolism, myocarditis, pericarditis, arrhythmias, heart failure	S	T
8	Electrolyte homeostasis	Nephrotic syndrome, respiratory disease, valvular disease	T	S
9	Renal failure	Electrolyte homeostasis, peripheral artery disease, pulmonary embolism, myocarditis, pericarditis, arrhythmias	S	T
10	Anaemia	Renal failure, nephrotic syndrome, respiratory disease	T	S
11	Lymphoma	Anaemia, electrolyte homeostasis, peripheral artery disease, pulmonary embolism, myocarditis, pericarditis	S	T
12	Solid tumours	All (exit examination)	T	T

T = test cases (cases contained key feature questions); S = study-only cases (cases did not contain key feature questions).

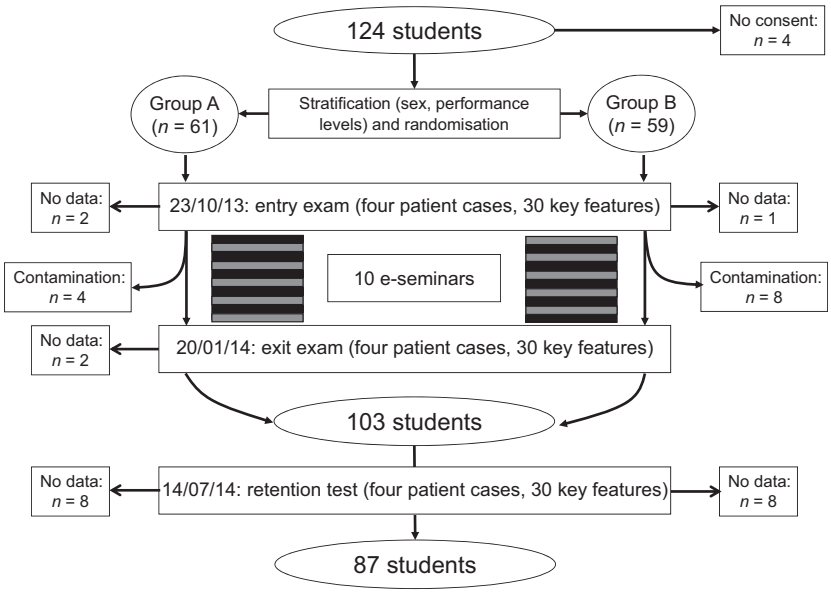

Figure 1 Flow of participants through the study. Black stripes = test cases; grey stripes = study-only cases. Contamination occurred when students were erroneously exposed to the wrong presentation format (by reporting to the computer room assigned to the other group) at least once

stripes). The presentation format was then alternated systematically for each group across 10 sessions. Study-only cases yielded exactly the same information as test cases; the only difference between them was the presence of five questions ('items') in each of the latter. Thus, each student answered a total of 100 items during the five intervention e-seminars to which he or she was assigned. Of these, 15 occurred in two e-seminars, which allowed for repeated testing of these 'intervention items'. As content was mutually exclusive for the two groups, intervention items for group A were never presented as questions in group B (and vice versa). Thus, by default, intervention items in group A served as control items in group B. Any student – regardless of group assignment – was exposed to 15 intervention items in two e-seminars and to 15 control items in two different e-seminars. The content of these 30 items had been pre-specified based on item characteristics obtained in a pilot phase in the preceding term. They were compiled into four cases containing six or nine key features each (with random allocation of items to cases); the diseases covered are presented in Table 2. In order to ensure comparability, the same cases were used in the entry and exit examinations, as well as the retention test, but they differed from the cases presented in e-seminars occurring between the entry and exit examinations. All examinations were formative in nature (i.e. no incentive to achieve high scores was provided as this may have confounded results[11]).

Detailed feedback was created for each item; students were able to access this feedback by clicking on a specific button. The same information provided in the feedback to specific questions was also provided as 'background information' in the control condition. Feedback for intervention items contained all options from the key feature long menu that were considered 'correct'. It also contained explanations of why some distractor options anticipated to be chosen by students were deemed 'incorrect'. The presentation of the information differed between groups in that feedback for intervention items was tailored to the question, whereas all relevant information was included in the 'background information' without highlighting those aspects that were tested in the intervention group. Content covered in e-seminars was not repeated during formal teaching and thus any increase in knowledge from the first intervention e-seminar to the exit examination will reflect exposure to the content during the e-seminar or self-study.

Student enrolment, data collection and analysis

Four weeks before the start of the winter term of 2013–2014, students were informed about the study by e-mail. On the first day of term, they were invited to provide written consent to participate in the study. Consenting students were then followed up over a total of 9 months. All three examinations, as well as all 10 e-seminars, were held in the institution's

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Table 2 Proportions of correct answers for the 30 key feature items in the exit examination and retention test

		Exit examination Condition		Retention test Condition	
Disease	**Key features**	**T (%)**	**S (%)**	**T (%)**	**S (%)**
Pulmonary embolism	Diagnosis of pulmonary embolism	59.5	48.9	57.1	64.4
	Wells score before D-dimer testing	76.2	60.0	76.2	51.1*
	Thorax CT scan to confirm PE	42.9	55.6	26.2	44.4
	Right ventricular strain for risk stratification	45.2	22.2*	42.9	24.4
	Fibrinolysis for unstable pulmonary embolism	59.5	51.1	54.8	37.8
Arterial hypertension	Diagnosis of secondary hypertension	40.5	20.0*	35.7	24.4
	Diagnosis of diastolic dysfunction	21.4	20.0	23.8	17.8
	Diagnosis of ACE inhibitor cough	88.1	82.2	73.8	86.7
Hyponatraemia	Hospital admission for hyponatraemia	31.0	6.7*	35.7	26.7
	Thiazide diuretics as cause of hyponatraemia	78.6	60.0	59.5	60.0
	Diagnosis of central pontine myelinolysis	40.5	17.8*	31.0	15.6
Atrial fibrillation	Orthostatic challenge after syncope	54.8	46.7	61.9	55.6
	ECG diagnosis of tachyarrhythmia	66.7	66.7	64.3	62.2
	CHA_2DS_2-VASc score for anticoagulation	38.1	33.3	64.3	46.7
	Brain CT scan for suspected stroke	71.1	53.3	54.8	71.1
Lupus erythematodes	Diagnosis of nephrotic syndrome	75.6	59.5	62.2	57.1
	Diagnosis of systemic lupus erythematodes	71.1	59.5	75.6	64.3
	Renal biopsy to confirm lupus nephritis	71.1	59.5	71.1	69.0
	Immunosuppressive treatment for lupus nephritis	71.1	64.3	71.1	73.8
COPD	Diagnosis of COPD	66.7	76.2	75.6	71.4
	Confirmation of COPD by $FEV_1/VC < 70\%$	68.9	35.7*	51.1	28.6*
	ABG analysis for suspected CO_2 intoxication	42.2	14.3*	48.9	21.4*
	Treatment of CO_2 intoxication by NIV	55.6	45.2	53.3	42.9
Pneumonia	Diagnosis of pneumonia in a chest X-ray	71.1	64.3	60.0	57.1
	CRB-65 score for hospital admission	53.3	28.6*	42.2	21.4*
Hyperthyroidism	Diagnosis of hyperthyroidism from lab results	86.7	71.4	82.2	81.0
	Stopping amiodarone in a patient with hyperthyroidism	57.8	40.5	51.1	40.5
Pulmonary fibrosis	Diagnosis of pulmonary fibrosis	53.3	38.1	44.4	42.9
	Amiodarone as cause of pulmonary fibrosis	57.8	54.8	75.6	71.4
	Indication for long-term oxygen therapy	57.8	50.0	48.9	35.7

* $p < 0.05$ in a chi-squared test comparing retesting and restudying. ABG = arterial blood gases; ACE = angiotensin-converting enzyme; CO_2 = carbon dioxide; COPD = chronic obstructive pulmonary disease; CRB = confusion/respiratory rate/blood pressure; CT = computed tomography; ECG = electrocardiogram; FEV_1 = forced expiratory volume in 1 second; NIV = non-invasive ventilation ; PE = pulmonary embolism; S = study-only cases (cases did not contain key feature questions); T = test cases (cases contained key feature questions); VC = vital capacity.

e-learning resource centre. Each student was assigned to one of the two computer rooms at the resource centre, and the presentation format (test cases or study-only cases) was switched for each room on a weekly basis. Data for students not reporting to the same room for every e-seminar were excluded from the analysis as these students received a higher or lower dose of the intervention, thus contaminating results. Examinations and e-seminars were scheduled to last no longer than 45 minutes within which students were free to complete sessions at their own pace. Individual login times were recorded.

The primary outcome for this study was the within-subject difference in percentage scores in intervention versus control items in the retention test at 6 months after the last e-seminar (paired *t*-test). Pearson correlations were calculated to assess the moderating effect of exposure time to study and test material on retention test performance. In addition, linear regression analyses were run using the difference between scores on test and study-only items in the exit and retention tests as the dependent variables, and the difference in mean time spent on test and study-only cases, respectively, as the independent variable, adjusting for student sex and percentage scores in summative end-of-module examinations as a surrogate parameter for overall performance levels. Secondary outcomes included student performance in the exit examination and the proportion of correct answers on individual items in both examinations.

Statistical analysis was performed using IBM SPSS Statistics for Windows Version 22.0 (IBM Corp., Armonk, NY, USA). Data are presented as the mean ± standard deviation (SD) or as numbers and percentages unless otherwise stated. A p-value of < 0.05 was considered to indicate statistical significance. This study was approved by the local ethics committee (application no. 2/10/13).

RESULTS

Student characteristics

The flow of participants through the study is displayed in Fig. 1. Of the 124 students eligible for study participation, four did not provide written consent. Following the exclusion of students for reasons of missing data or contamination (see above), complete data were available for 87 students (longitudinal sample; effective response rate: 70.2%). The mean ± SD age of the study participants was 25.0 ± 2.9 years and 58.6% (n = 51) were female. There were no significant differences regarding age, sex distribution and previous examination scores between students in the longitudinal sample and the 33 students excluded from the analysis (data not shown).

Descriptive analysis of e-seminars and examinations

Attendance at e-seminars was high: no student missed more than two seminars. Mean login time at e-seminars was 17.46 ± 4.44 minutes, and students spent significantly more time on test items than on study-only items (21.47 ± 5.27 minutes versus 13.30 ± 5.24 minutes per seminar; p < 0.001). The percentage of points scored on test items ranged from 38.6% to 65.6%, reflecting the absence of a floor or ceiling effect. The internal consistency of e-seminars (test items) tended to increase over the course of the study with the final four sets of test seminars (20 items each) all yielding Cronbach's α-values of > 0.8. Cronbach's α-values for the entry, exit and retention examinations were 0.663, 0.905 and 0.895, respectively.

Learning outcome

Mean ± SD scores in the entry, exit and retention examinations were 22.6 ± 11.3%, 53.0 ± 24.4% and 52.4 ± 23.4%, respectively. Figure 2 shows superior performance on test items compared with study-only items by the primary measure of long-term retention. Mean ± SD scores for test items and study-only items were 56.0 ± 25.8% and 48.8 ± 24.7%, respectively (p < 0.001). The difference between the two scores was slightly larger on the secondary measure of exit examination performance (59.3 ± 27.7% versus 46.7 ± 24.8%; p < 0.001). There was a significant correlation between extra time spent on test items and the difference between performance on test and study-only items in the retention test (r = 0.273, p = 0.011), but not in the exit examination (r = 0.038, p = 0.724). After adjusting for student sex and performance in summative end-of-course examinations, the main effect of the intervention on examination performance remained significant for both the retention test (intercept: -7.03, 95% confidence interval [CI] -12.03 to -2.03; p = 0.006) and the exit examination (intercept: -8.40, 95% CI -13.71 to -3.09; p = 0.002).

Student performance on individual items in the exit and retention tests is given as a function of prior exposure (repeated testing versus repeated study) in Table 2. Performance trajectories (Fig. 3) indicate that once an intervention item had been answered correctly, another correct answer was provided on a subsequent test in 72.0% of cases. More importantly, 33.3% of students who failed to provide a correct answer in one test went on to submit a correct answer in a subsequent test of the same item, thus suggesting a learning effect elicited by the feedback provided.

DISCUSSION

To our knowledge, this is the first study using repeated key feature examinations in a test-

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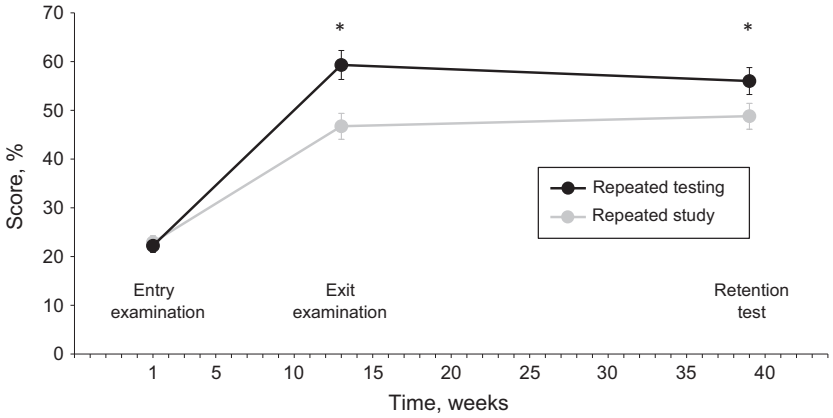

Figure 2 Learning outcomes. Student performance in the entry and exit examinations and the retention test for test items (black line) and study-only (grey line) items, respectively. Error bars indicate standard errors of the mean. *p < 0.001 in a paired *t*-test comparing student performance on intervention and control items

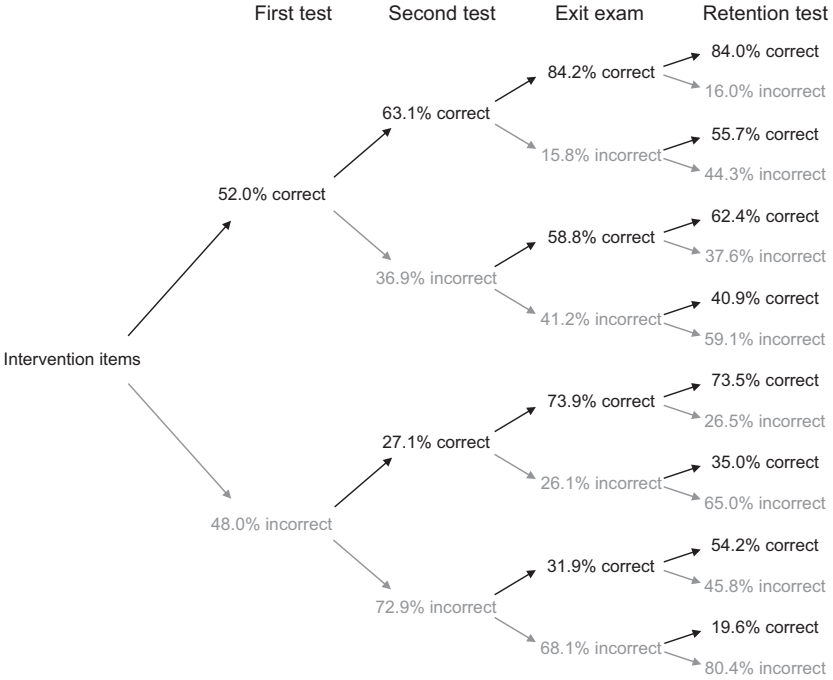

Figure 3 Performance trajectories showing the proportions of intervention items answered correctly as a function of prior performance on identical items. Proportions of each category are connected by arrows from the preceding test

enhanced learning paradigm. It is also the first study to specifically address clinical reasoning in undergraduate medical students. After 6 months, we found a sustained effect of repeated testing that was robust even when additional exposure time to the material in testing sessions was accounted for. The key feature examinations used in our study covered a wide range of diseases. Students who had been repeatedly tested outperformed students who had repeatedly studied the same material on tasks such as diagnosing pneumonia on a chest X-ray, detecting bronchial obstruction in a lung function test and calculating the Wells score before ordering a D-dimer test in suspected pulmonary embolism, all of which are highly relevant to clinical practice. Once established, key feature testing appears to be a time-efficient way of enhancing retention of clinical reasoning. For some learning objectives, it may thus be an intriguing alternative to more resource-intensive teaching formats.

Comparison with previous research

Previous studies on the effectiveness of test-enhanced learning in medical education differ with regard to study design (observational[23] or randomised[24]), study population (medical,[25] dental[15] and nursing[17] students), career level (from incoming students[23] to residents[26]), sample size (21[23] to 138[27]), learning objectives (from factual knowledge[14] to clinical applications[26]), testing format (from multiple-choice questions[17] to essays[25] and simulated patient encounters[16]), number of interventions (one,[14] three[26] or four[25]), and length of follow-up (1 week[17] to 6 months[16]). Although this huge variation limits the extent to which our results can be compared with those of earlier studies, there are some similarities between the present study and recent reports by Larsen and colleagues,[16,25,26,28] who studied test-enhanced learning in neurology. In one randomised crossover study, 40 neurology residents who had received teaching on myasthenia gravis and status epilepticus subsequently participated in three testing or studying sessions over a period of 4 weeks. Six months later, performance on short-answer questions was better on items that had been repeatedly tested (scores: 39% versus 26%). In another crossover trial conducted by the same group,[25] 47 Year 1 medical students received teaching on seizures, optic neuritis, myasthenia gravis and migraine. In a retention test taken 6 months later, repeated testing was associated with significantly higher percentage scores than repeated study (40% versus 20%).[25] Similar results were obtained in a more recent trial involving sophisticated methods, in which the authors used simulated patients for intervention sessions and the final retention test.[16] Although this approach demonstrates that the effects elicited by test-enhanced learning may translate into clinical practice, it also raises the issue of resource allocation within medical education.[28] Repeated testing with simulated patients is time-consuming and requires considerable staff numbers. Given the economies of scale associated with e-learning interventions[29] and the educational rationale for using key feature questions to foster complex cognitive functions,[12] our study contributes to the growing body of literature on how e-learning can be used efficiently to improve student learning outcomes.[30] Repeated key feature examinations can create learning opportunities that otherwise might not be available to all students (e.g. performing an arterial blood gas analysis in a case of suspected carbon dioxide intoxication [Table 2]).

At first glance, the overall performance in the retention test, which resulted in a mean score of just over 50%, is disappointing. However, the percentage scores achieved for intervention and control items compare favourably with those in earlier studies.[25,26] The results are likely to reflect the fact that the retention test was formative in nature. As these assessments did not generate credit points, students did not revise for them, which avoided any potential confounding imposed by the impact of examination consequences on learning behaviour. Accordingly, our data are likely to reflect the true effect of repeated testing on the ability to access the knowledge needed to solve the clinical problems presented.

One unexpected finding of the present study was the absence of a substantial decline in examination performance over a period of 6 months. Few studies on test-enhanced learning in medical education have specifically reported on retention, but those that have done suggest a considerable decrease in performance levels over the course of 6 months.[14,16,25,26] As in most other studies,[14,17,26] the exit examination and retention test were identical. Consequently, one potential explanation for the lack of a performance loss is that students may simply have remembered the cases, questions and answers. This is unlikely because the students were not reminded of the content over the period of 6 months. The magnitude of this effect would be expected to be similar for intervention and control items, and even if this effect did have an impact on our results, it was too weak to mask the sustained advantage of testing over studying. An alternative explanation for the excellent retention observed in this study is that students applied their knowledge in consecutive clinical attachments. This – and the fact that repeated study appeared to elicit superior retention in some items (see Table 2) – highlights the need for a multifaceted approach to teaching that blends traditional teaching formats with innovative concepts.

Strengths and limitations of the study

This study was designed according to current recommendations for this type of research.[12] We used a production test format rather than a recognition test format to assess clinical reasoning. Testing was spaced across weeks and students were given immediate educational feedback on their answers. The response rate was acceptable and there was no evidence of selection bias threatening the validity of

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the findings. The internal consistency of the exit and retention tests was excellent. This study was not designed to address the cognitive mechanisms underlying the observed effect. However, our results are in line with the hypothesis that repeated testing with key feature questions enhances knowledge retention. Given the crossover design and the within-subjects comparison of intervention and control items, our data suggest that the observed score difference can be attributed to an effect of repeated testing on the storage and retrieval of knowledge rather than an effect of training in how to answer key feature questions. The latter is unlikely because all students received the same 'dose' of key feature training. Unlike another recent study,[21] we found an overall effect of repeated testing despite the considerable differences between the cases presented in the exit examination and the cases presented during e-seminars, although both sets targeted identical learning objectives. We cannot comment on the cognitive processes underlying clinical reasoning that were primarily invoked by our intervention. It may be hypothesised that the cases were easy enough to be solved using an intuitive strategy, thus requiring less higher-order metacognitive activity.[1] This hypothesis should be tested in future trials.

Despite these strengths, the generalisability of our findings is limited by the monocentric nature of the study. As this study was meant to shed light on the real-world effectiveness of test-enhanced learning, some potential confounding factors were not experimentally controlled. Most importantly, we did not collect data on independent student learning. Test cases might have stimulated self-study outside the e-learning resource centre to a greater extent than study-only cases, thus magnifying the observed effect. Unlike laboratory studies on test-enhanced learning, we did not attempt to control time-on-task during e-seminars but allowed students to terminate their sessions at their own convenience. However, login times were recorded and included in our analysis. There was weak evidence of an association between extra time-on-task in test sessions and better learning outcomes on test items, but the effect of the intervention persisted when this was controlled for in the analysis. Finally, this study did not assess whether repeated testing with key feature questions impacts on student performance in the clinical setting. Although one study suggests such a link,[16] more research is needed to establish a causal relationship between repeated testing and more favourable patient outcomes.

CONCLUSIONS

For the first time, this study demonstrates that repeated case-based learning augmented with formative key feature questions is more effective than case-based learning alone in fostering clinical reasoning in undergraduate medical students. The effect was sustained over 6 months and cannot be explained by differences in exposure time between the two presentation formats. Repeated testing using key feature questions appears to be a time-efficient way of enhancing retention of procedural knowledge in undergraduate medical education.

Contributors: TR conceived the study, developed its design, drafted key feature cases, analysed the data and wrote the manuscript. JCA implemented all key feature examinations on the learning platform and contributed to data analysis and the writing of the paper. KM supervised all key feature examinations in the e-learning resource centre, was responsible for data handling and prepared the statistical analysis. She also contributed to the writing of the Methods and Discussion sections. LS wrote all study-only cases and drafted feedback narratives for all 200 key feature questions. She also contributed to the writing of the Methods and Discussion sections. MK drafted all key feature cases related to nephrology and rheumatology and revised these cases following the pilot phase in the preceding term. He contributed to the writing of the Results and Discussion sections. WJ helped to write all key feature cases related to haematology and oncology and revised these cases following the pilot phase. He contributed to the writing of the Results section. JB designed major aspects of the statistical analysis plan, helped analyse the data, and contributed to the Introduction, Results and Discussion sections. SA was involved in the study at the design stage, specifically contributed to the design of the three major examinations (entry, exit, retention), and contributed to the Introduction and Discussion sections. All authors contributed to the critical revision of the paper and approved the final manuscript for submission.

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Conflicts of interest: none.
Ethical approval: this study was approved by the Göttingen University Medical Centre Ethics Committee (no. 2/10/13).

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